

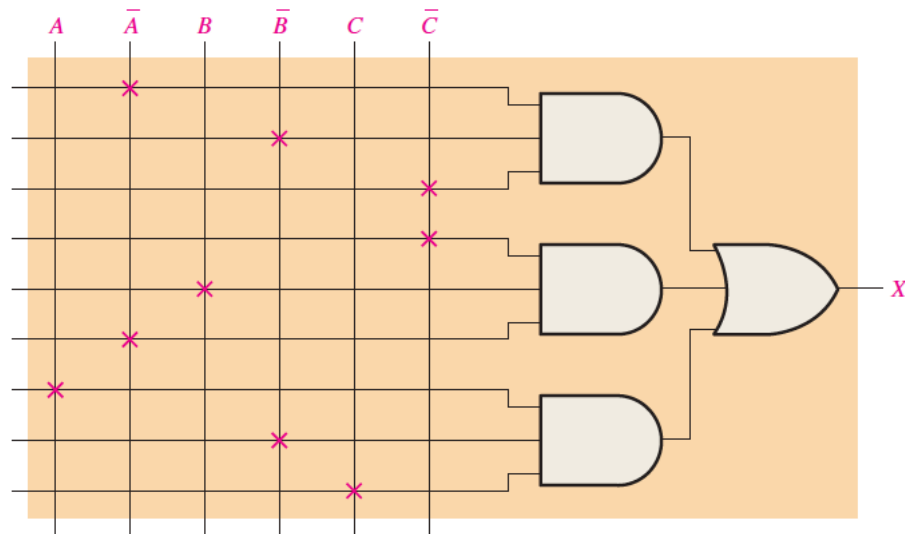
Digital Logic Circuits (Spring 2026)

Quiz #4

Name: _____ Student ID: _____

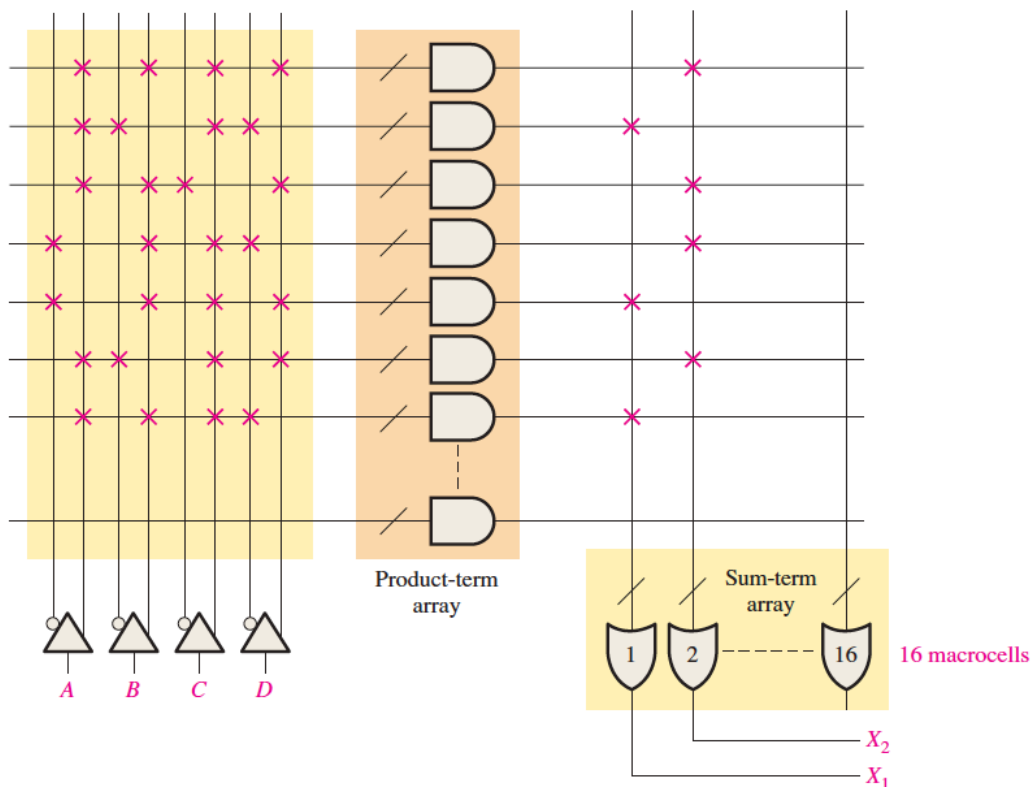
1. A Moore state machine consists of combinational logic circuits that determine **C**
 - a) sequences
 - b) memory
 - c) both (a) and (b)
 - d) neither (a) nor (b)
2. The output of a Mealy machine depends on its **A**
 - a) inputs
 - b) next state
 - c) present state
 - d) answers (a) and (c)
3. The maximum cumulative delay of an asynchronous counter must be **B**
 - a) More than the period of the clock waveform
 - b) Less than the period of the clock waveform
 - c) Equal to the period of the clock waveform
 - d) Both (a) and (c)
4. Two types of SPLDs are **C**
 - a) CPLD and PAL
 - b) PAL and FPGA
 - c) PAL and GAL
 - d) GAL and SRAM
5. The basic elements of an FPGA are **D**
 - a) Configurable logic blocks
 - b) I/O blocks
 - c) PAL arrays
 - d) Both (a) and (b)
6. The final output of the synthesis phase of a design flow is the **A**
 - a) netlist
 - b) bitstream
 - c) timing simulation
 - d) device pin numbers
7. Data that are stored at a given address in a RAM are lost when **D**
 - a) Power goes off
 - b) The data are read from the address
 - c) New data are written at the address
 - d) Answers (a) and (c)

8. Determine the Boolean output expression for the simple PAL array shown in Figure below. The Xs represent connected links.



$$X = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C$$

9. Determine the output expressions for X_1 and X_2 from macrocells 1 and 2 in Figure below.



$$X_1 = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}BCD + ABC\bar{D}; \quad X_2 = ABCD + ABC\bar{D} + \bar{A}BC\bar{D} + \bar{A}\bar{B}CD$$

10. Please list all the memory types you can remember (no less than five).

SRAM, DRAM, EPROM, EEPROM, FLASH, Hard disk, and any other things reasonable